

VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.05)

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1. EXECUTIVE SUMMARY

This report documents the validation of the Shielding Attenuation Simulator V1.05. Version 1.05 expands the Geometric Progression (G-P) buildup factor implementation from single layer lead shielding to multiple supported shielding materials.

Version 1.04 introduced and validated the G-P buildup calculation framework using lead. That earlier validation confirmed coefficient interpolation, buildup factor calculation, buildup-corrected flux calculation, and rejection of unsupported buildup cases. Version 1.05 builds on that validated framework by adding G-P exposure buildup coefficient tables for additional materials and verifying that each supported material produces physically consistent buildup behavior.

The V1.05 implementation supports G-P buildup correction for single layer homogeneous shields made from:

- Lead
- Aluminum
- Ordinary Concrete
- Water
- Tungsten
- Copper
- Tin

Buildup still unsupported:

- Barite Concrete
- Polyethylene

Multilayer buildup remains intentionally unsupported because applying a single material buildup factor to mixed shielding layers may produce misleading results. Barite concrete and polyethylene remain unsupported because G-P coefficient tables for those simulator materials have not been added or validated in V1.05.

2. METHODOLOGY AND PHYSICS MODELS

2.1 Baseline Attenuation and Flux Model

The primary calculation for shielding effectiveness remains unchanged from V1.02 and V1.04 and utilizes the Beer-Lambert Law. The simulator calculates the transmission (T) of a mono-energetic photon beam through N layers of material:

$$T = \exp\left(-\sum_{i=1}^N \mu_i x_i\right)$$

where:

- μ_i is the linear attenuation coefficient of the i -th material (cm^{-1}).
- x_i is the thickness of the i -th layer (cm).
- T is the dimensionless narrow-beam transmission.

2.2 Material Data and Interpolation

Attenuation coefficients are derived from NIST XCOM mass attenuation data (μ/ρ). For energies between tabulated data points, the simulator employs log-log interpolation to account for the power-law dependence of attenuation coefficients on energy.

$$\ln(\mu) = \ln(\mu_1) + \frac{\ln(E) - \ln(E_1)}{\ln(E_2) - \ln(E_1)} \cdot (\ln(\mu_2) - \ln(\mu_1))$$

2.3 Geometric Flux Calculation

The uncollided detector flux (ϕ) is calculated by combining the shield transmission with geometric spreading from a point source:

$$\phi_{\text{uncollided}} = \frac{S \cdot T}{4\pi r^2}$$

where:

- S is the source strength in photons / s
- r is the distance from the source to the detector (cm).
- T is the dimensionless shield transmission.

2.4 G-P Buildup Factor Model

Version 1.05 uses the same methods from V1.04 which added an optional buildup correction:

$$\phi_{\text{buildup}} = \phi_{\text{uncollided}} \cdot B$$

where:

- $\phi_{\text{uncollided}}$ is the narrow-beam uncollided flux
- B is the G-P buildup factor

- $\phi_{buildup}$ is the buildup-corrected flux estimate

The buildup factor is calculated using G-P fitting coefficients:

$$B = 1 + \frac{b-1}{K-1} (K^x - 1)$$

for:

$$K \neq 1$$

where:

- B is the buildup factor
- x is shield thickness in mean free paths
- b is a G-P fitting coefficient
- K is an intermediate G-P fitting value

The value of K is calculated as:

$$K = cx^a + d \cdot \frac{\tanh(\frac{x}{X_k}-2) - \tanh(-2)}{1 - \tanh(-2)}$$

where:

- x is the shield thickness in mean free paths
- c , a , X_k and d are G-P fitting coefficients
- $\tanh$ is the hyperbolic tangent function

When $K = 1$, the normal G-P buildup equation would involve division by zero. In this case, the simulator uses the limiting form:

$$B = 1 + (b - 1)x$$

for:

$$K = 1$$

where:

- B is the buildup factor
- x is shield thickness in mean free paths
- b is a G-P fitting coefficient
- K is an intermediate G-P fitting value

In the implementation, the simulator checks whether K is extremely close to 1.0 using a numerical tolerance. If:

$$|K - 1| < 10^{-12}$$

the limiting form is used instead of the standard G-P expression.

The implemented coefficient order is

$$material\ name, energy, b, c, a, X_k, d$$

The current V1.05 implementation supports G-P buildup correction for single-layer homogeneous shielding cases using lead, aluminum, ordinary concrete, water, tungsten, copper, and tin. Multilayer buildup remains intentionally unsupported.

2.5 Coefficient Source

The lead G-P coefficients used in V1.05 are from D. K. Trubey's ORNL/RSIC--49/R1; NUREG/CR-5740 report, *New Gamma-Ray Buildup Factor Data for Point Kernel Calculations: ANS-6.4.3 Standard Reference Data*. The OSTI record identifies the report as a 1991 Oak Ridge National Laboratory technical report and describes the ANS-6.4.3 evaluated buildup factor database for point-kernel calculations.

2.6 Material Key Lookup

Version 1.05 adds an internal material key to the Material model. This allows the simulator to use readable material display names while still using stable dictionary keys for coefficient lookup. The change prevents display-name formatting from interfering with G-P coefficient lookup.

3. TEST CASE RESULTS

3.1 G-P Buildup Identity Tests

The validation runner checks each supported material using two known G-P buildup identities:

- $B(0\text{ MFP}) = 1$
- $B(1\text{ MFP}) = b$

The zero MFP test confirms that the buildup factor correctly returns 1.0 when no shielding thickness is present. The 1 MFP test confirms the G-P equation behavior where the buildup factor equals the b coefficient at one mean free path.

Test Case	Input Parameters	Expected Result	Actual Result	Status
1	Lead, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
2	Lead, 1 MFP	$B = b$	$B = b$	PASS

3	Aluminum, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
4	Aluminum, 1 MFP	$B = b$	$B = b$	PASS
5	Ordinary Concrete, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
6	Ordinary Concrete, 1 MFP	$B = b$	$B = b$	PASS
7	Water, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
8	Water, 1 MFP	$B = b$	$B = b$	PASS
9	Tungsten, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
10	Tungsten, 1 MFP	$B = b$	$B = b$	PASS
11	Copper, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
12	Copper, 1 MFP	$B = b$	$B = b$	PASS
13	Tin, 0 MFP	$B = 1.0$	$B = 1.0$	PASS
14	Tin, 1 MFP	$B = b$	$B = b$	PASS

The zero-MFP test confirms that the buildup factor correctly returns 1.0 when no shielding thickness is present. The 1-MFP test confirms the G-P equation behavior where the buildup factor equals the b coefficient at one mean free path.

3.2 Finite Shield Thickness Buildup Tests

Test Case	Input Parameters	Expected Result	Actual Result	Status
15	5 cm, 0.6617 MeV, $S = 3.7 \times 10^{10}$ photons/s, $r = 100$ cm	$\Phi_{buildup} \approx$ 1.42612×10^3 photons/cm ² /s	$\Phi_{buildup} \approx$ 1.426119×10^3 photons/cm ² /s	PASS

For this test case, the previously validated narrow-beam model gives:

$$T \approx 0.0018362592$$

$$\Phi_{uncollided} \approx 5.40662 \times 10^2 \text{ photons/cm}^2/\text{s}$$

At 0.6617 MeV through 5 cm of lead:

$$x \approx 6.30002 \text{ MFP}$$

$$B \approx 2.63773$$

Therefore:

$$\phi_{buildup} = \phi_{uncollided} \cdot B$$

$$\phi_{buildup} \approx 1.426119 \times 10^3 \text{ photons/cm}^2/\text{s}$$

The result is physically consistent because the buildup corrected flux is greater than the uncollided narrow-beam flux.

3.3 Error Handling and Unsupported Case Tests

Test Case	Input Parameters	Expected Result	Actual Result	Status
16	Apply buildup to barite concrete.	ValueError: G-P buildup mode does not currently support Concrete (Barite, Type BA).	ValueError: G-P buildup mode does not currently support Concrete (Barite, Type BA).	PASS
17	Apply buildup to 2 layers.	ValueError: G-P buildup mode currently supports only one shielding layer.	ValueError: G-P buildup mode currently supports only one shielding layer.	PASS
18	Request lead G-P coefficients at 0.01 MeV.	ValueError: Requested energy is outside the G-P coefficient data range.	ValueError: Requested energy is outside the G-P coefficient data range.	PASS
19	Pass -1 cm layer thickness.	ValueError: Layer thickness cannot be negative.	ValueError: Layer thickness cannot be negative.	PASS
20	Total shielding thickness greater than detector distance.	ValueError: Total shielding thickness cannot be greater than the source-to-detector distance.	ValueError: Total shielding thickness cannot be greater than the source-to-detector distance.	PASS

3.4 Finite Shield Thickness Buildup Tests

The validation runner also checks that each supported material produces a buildup factor greater than 1.0 at 5 mean free paths.

Test Case	Input Parameters	Expected Result	Actual Result	Status
21	Lead, 5 MFP	$B > 1.0$	$B > 1.0$	PASS
22	Aluminum, 5 MFP	$B > 1.0$	$B > 1.0$	PASS
23	Ordinary Concrete, 5 MFP	$B > 1.0$	$B > 1.0$	PASS
24	Water, 5 MFP	$B > 1.0$	$B > 1.0$	PASS
25	Tungsten, 5 MFP	$B > 1.0$	$B > 1.0$	PASS
26	Copper, 5 MFP	$B > 1.0$	$B > 1.0$	PASS
27	Tin, 5 MFP	$B > 1.0$	$B > 1.0$	PASS

The 5 MFP test provides a basic check that the coefficient tables produce physically reasonable buildup behavior for nonzero shielding thickness.

3.5 Material Key Lookup Test

Version 1.05 changes the material model so that each material has both a display name and an internal key. This is especially important for ordinary concrete because its display name contains parentheses and spaces.

Test Case	Input Parameters	Expected Result	Actual Result	Status
28	Select concrete_ordinary and apply buildup	G-P coefficients found using material key.	G-P coefficients found using material key.	PASS

This confirms that the simulator can use the readable display name Concrete (Ordinary) while still finding the coefficient table stored under concrete_ordinary.

4. ASSUMPTIONS AND LIMITATIONS

1. Single Layer Homogeneous Buildup Only: V1.05 supports G-P buildup correction only for single layer homogeneous shielding cases.
2. No Multilayer Buildup Treatment: Multilayer buildup is not implemented. The simulator rejects buildup mode for multiple shielding layers because applying a single material buildup factor to mixed materials may produce misleading results.
3. Supported Buildup Materials: V1.05 supports G-P exposure buildup correction for lead, aluminum, ordinary concrete, water, tungsten, copper, and tin.

4. Unsupported Buildup Materials: Barite concrete and polyethylene remain unsupported for buildup correction in V1.05, although they remain available for narrow-beam attenuation calculations.
5. Point-Source Geometry: The source is still modeled as an isotropic point source.
6. Monoenergetic Photon Source: The simulator currently models one photon energy per calculation.
7. Narrow-Beam Transmission Still Used as Baseline: The buildup-corrected flux is calculated by multiplying the narrow-beam uncollided flux by the G-P buildup factor.
8. Coefficient Interpolation: G-P coefficients are linearly interpolated between tabulated energies. This is separate from the NIST XCOM attenuation interpolation, which uses log-log interpolation.
9. Exposure Buildup Coefficients: The implemented coefficients are exposure buildup coefficients. Future versions may distinguish exposure, dose, and energy buildup factors if additional coefficient tables are added.
10. Simple 1D Geometry: The simulator assumes shielding thickness lies along the source to detector path. Therefore, total shielding thickness cannot exceed detector distance.

5. CONCLUSION

The Shielding Attenuation Simulator V1.05 successfully expands G-P buildup correction from lead-only support to multiple single-layer shielding materials. The simulator now supports G-P exposure buildup correction for lead, aluminum, ordinary concrete, water, tungsten, copper, and tin.

Validation tests confirmed that each supported material satisfies expected G-P buildup behavior at 0 and 1 mean free paths. Additional finite-thickness tests confirmed that each supported material produces a buildup factor greater than 1.0 at 5 mean free paths. The simulator also continues to reject unsupported cases, including multilayer buildup, materials without G-P coefficient tables, and physically invalid source-to-detector geometry.

V1.05 represents a meaningful improvement over V1.04 by extending the validated G-P buildup framework to multiple shielding materials while preserving the intentional limitation to

single-layer homogeneous shielding. Future work should focus on improved input handling, dose-rate conversion, isotope libraries, additional buildup coefficient types, and eventual comparison against Monte Carlo transport tools.

References

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